

Review Article



Integrative Roles of NK Receptors and $\gamma\delta$ TCRs in $\gamma\delta$ T Cell-Mediated Cancer Immunity

Won Joon Oh ^{1,2}, Haryoung Cho ^{1,2}, Tilanka Kalanapriya Dahanayake ^{1,2},
Tae Jin Kim ^{1,2,*}

¹Department of Immunology, Sungkyunkwan University School of Medicine, Suwon 16419, Korea

²Personalized Cancer Immunotherapy Research Center, Sungkyunkwan University Medical Research Center, Suwon 16419, Korea



Received: Dec 5, 2025

Revised: Jan 9, 2026

Accepted: Jan 14, 2026

Published online: Feb 10, 2026

*Correspondence to

Tae Jin Kim

Department of Immunology, Sungkyunkwan University School of Medicine, 2066 Seobu-ro, Jangan-gu, Suwon 16419, Korea.
Email: tjkim@skku.edu

Copyright © 2026. The Korean Association of Immunologists

This is an Open Access article distributed under the terms of the Creative Commons Attribution Non-Commercial License (<https://creativecommons.org/licenses/by-nc/4.0/>) which permits unrestricted non-commercial use, distribution, and reproduction in any medium, provided the original work is properly cited.

ORCID iDs

Won Joon Oh

<https://orcid.org/0009-0008-9250-2513>

Haryoung Cho

<https://orcid.org/0009-0002-5080-0084>

Tilanka Kalanapriya Dahanayake

<https://orcid.org/0009-0000-1553-3596>

Tae Jin Kim

<https://orcid.org/0000-0001-9802-0568>

Conflict of Interest

The authors declare no potential conflicts of interest.

ABSTRACT

$\gamma\delta$ T cells employ dual recognition strategies that integrate innate and adaptive immune sensing through NK receptors (NKR) and $\gamma\delta$ -TCRs in cancer immunity. Through the coordinated use of clonotypic $\gamma\delta$ -TCRs and germline-encoded NKRs, $\gamma\delta$ T cells recognize malignant cells in an MHC-unrestricted manner and exert potent cytotoxic and immunomodulatory functions. In this review, we discuss how activating and inhibitory NKRs—including NKG-2D, natural cytotoxicity receptors, CD16, NKG-2A, and killer cell immunoglobulin-like receptors—are expressed across major human $\gamma\delta$ T cell subsets and how their signals are integrated with $\gamma\delta$ -TCRs. We highlight subset-specific differential utilization of NKRs in circulating V δ 2 and tissue-resident V δ 1 $\gamma\delta$ T cells, context-dependent integration of $\gamma\delta$ -TCR and NKR signaling within the tumor microenvironment, and emerging implications for $\gamma\delta$ T cell-based cancer immunotherapy.

Keywords: T lymphocyte subsets; Receptors, antigen, T-cell, gamma-delta; Receptors, natural killer cell; Cancer

INTRODUCTION

Although $\gamma\delta$ T cells represent a minor subset of T cells, they play critical roles in tumor immunity through both innate and adaptive mechanisms (1). Similar to NK cells, $\gamma\delta$ T cells express various activating NK receptors (NKR), including NKG-2D and natural cytotoxicity receptors (NCR), which enable recognition of stress-induced ligands on tumor cells, and inhibitory NKRs such as NKG-2A and killer immunoglobulin-like receptors (KIRs). In addition, their clonotypic $\gamma\delta$ -TCRs allow for recognition of various non-peptide Ags and tumor-associated molecules, conferring a broader target range than conventional $\alpha\beta$ T cells (2). However, how signaling through NKRs and $\gamma\delta$ -TCRs is integrated to shape $\gamma\delta$ T-cell activation, effector function, and functional plasticity within the tumor microenvironment (TME) remains incompletely understood, despite the well-recognized pleiotropic roles of $\gamma\delta$ T cells in cancer (3).

Table 1. Representative activating and inhibitory NK cell receptors expressed on V δ 1 and V δ 2 $\gamma\delta$ T cells

	Receptor	Ligands	V δ 1	Activated V δ 1	V δ 2	Activated V δ 2	References
Activating NKRs	NKG-2D (CD314)	MICA/B, ULBPs	Very high 60%–90%	Very high 90%	High 60%–80%	Very high >90%	(7–9)
	NCR1 (Nkp46)	Viral HA/HN, heparan sulfate	Tissue-specific PBMC: <5% Gut IEL: 30%–60%	Low–moderate <20%	Very low <5%	Low 5%–10%	(9,10)
	NCR2 (Nkp44)	Viral HA, PDGF-DD, heparan sulfate	Very low <5%	Moderate 10%–25%	Very low <5%	Moderate 10%–30%	(7,9,11)
	NCR3 (Nkp30)	B7-H6, BAG6/BAT3, heparan sulfate	Low 5%–15%	Moderate 15%–30%	Very low <5%	Low 5%–15%	(9,11,12)
	DNAM-1 (CD226)	CD155 (PVR), CD112 (Nectin-2)	High 40%–60%	Very high >90%	Very high >80%	Very high >90%	(7,8)
	CD16 (Fc γ RIIA)	Fc portion of IgG (ADCC)	Low–moderate 5%–25%	High 40%–60%	Low 5%–15%	Moderate 15%–30%	(7,8,13)
Inhibitory NKRs	NKG2C (KLRC2)	HLA-E	Low 5%–15% (CMV+: 20%–40%)	Moderate 20%–40%	Very low <5%	Low 5%–20%	(14,15)
	NKG2A (KLRC1)	HLA-E	Very low <5%–10%	Very low <5%–10%	Moderate 20%–50%	Variable 20%–90%	(16)
	KIR2DL1/2/3	HLA-C (C1/C2 epitopes)	Low 5%–15%	Moderate 15%–30%	Very low <5%	Low 5%–15%	(15,17)
	KIR3DL1	HLA-Bw4, HLA-A	Low 5%–15%	Moderate 15%–30%	Very low <5%	Low 5%–10%	(15,17)
	KIR3DL3	HHLA2 (B7 family)	Low 5%–15%	Moderate 15%–30%	Very low <5%	Low 5%–10%	(18)

Categories of expression levels: very high (>80%, constitutive); high (50%–80%); moderate (20%–50%); low (5%–20%); very low (<5%); negative (<1%). Activated V δ 1 and V δ 2 cells *in vitro*: V δ 2 cells treated with zoledronate (1–5 μ M) + IL-2 (100–500 IU/mL) for 10–14 days; V δ 1 cells treated with anti-CD3/CD28 + IL-2/IL-15. HA/HN, hemagglutinin/hemagglutinin-neuraminidase; PVR, poliovirus receptor.

Abbreviations

ADCC, Ab-dependent cellular cytotoxicity; BTN, butyrophilin; CMV, cytomegalovirus; IPP, isopentenyl pyrophosphate; ITAM, immunoreceptor tyrosine-based activation motif; ITIM, immune-receptor tyrosine-based inhibition motif; KIR, killer immunoglobulin-like receptor; MICA/B, MHC I chain-related molecule A/B; NCR, natural cytotoxicity receptor; NK, NK receptor; pAg, phosphoantigen; TME, tumor microenvironment.

Author Contributions

Conceptualization: Oh WJ, Dahanayake TK, Kim TJ; Data curation: Oh WJ; Funding acquisition: Kim TJ; Investigation: Cho H, Dahanayake TK, Kim TJ; Project administration: Kim TJ; Supervision: Kim TJ; Writing - original draft: Oh WJ, Cho H, Dahanayake TK, Kim TJ; Writing - review & editing: Kim TJ.

The dual recognition system mediated by innate NKRs and adaptive $\gamma\delta$ -TCRs renders $\gamma\delta$ T cells potent effectors of anti-tumor immunity and promising candidates for next-generation immunotherapies. These receptors may act synergistically to enhance cytotoxic activity or compete for shared signaling pathways, resulting in context-dependent modulation of anti-tumor immune responses (4,5). A comprehensive understanding of how NKRs and $\gamma\delta$ -TCRs coordinate signaling and effector mechanisms is essential for optimizing $\gamma\delta$ T cell-based therapeutic strategies. This complexity is further increased by the diverse repertoire of activating and inhibitory NKRs expressed on $\gamma\delta$ T cells, which generates a dynamic regulatory network within the TME. V δ 1 and V δ 2 $\gamma\delta$ T cells express differential sets of activating and inhibitory NKRs (Table 1, Fig. 1). Although Table 1 lists only inhibitory KIRs, activating KIRs with opposing functions are discussed in the KIR section. The broader functional differences between V δ 1 and V δ 2 $\gamma\delta$ T cells have been extensively reviewed elsewhere and are therefore not discussed in detail here (1). The complexity of NKRs and $\gamma\delta$ -TCRs highlights the necessity for further research to elucidate the specific roles of these receptors and their interactions in $\gamma\delta$ T cell-mediated tumor immunity. Anti-tumor $\gamma\delta$ T cell responses may be not driven by a single dominant receptor system but rather by dynamic integration of $\gamma\delta$ -TCR and NKR signaling, shaped by tissue localization, ligand availability, and chronic stimulation within the TME. Moreover, depending on environmental cues, $\gamma\delta$ T cells may also acquire pro-tumor or immunoregulatory functions, including differentiation into Treg- or Th17-like cells and direct suppression of $\alpha\beta$ T-cell responses (6). In this review, we integrate current knowledge on the expression, regulation, and function of NKRs in $\gamma\delta$ T cells, with particular emphasis on their coordinated signaling with $\gamma\delta$ -TCRs in cancer immunity.

ACTIVATING NK CELL RECEPTORS EXPRESSED ON $\gamma\delta$ T CELLS

The sufficient activation of activating NKRs and/or $\gamma\delta$ -TCRs is critical for the anti-tumor function, but their activity may be inhibited by inhibitory NKRs discussed in the next section.

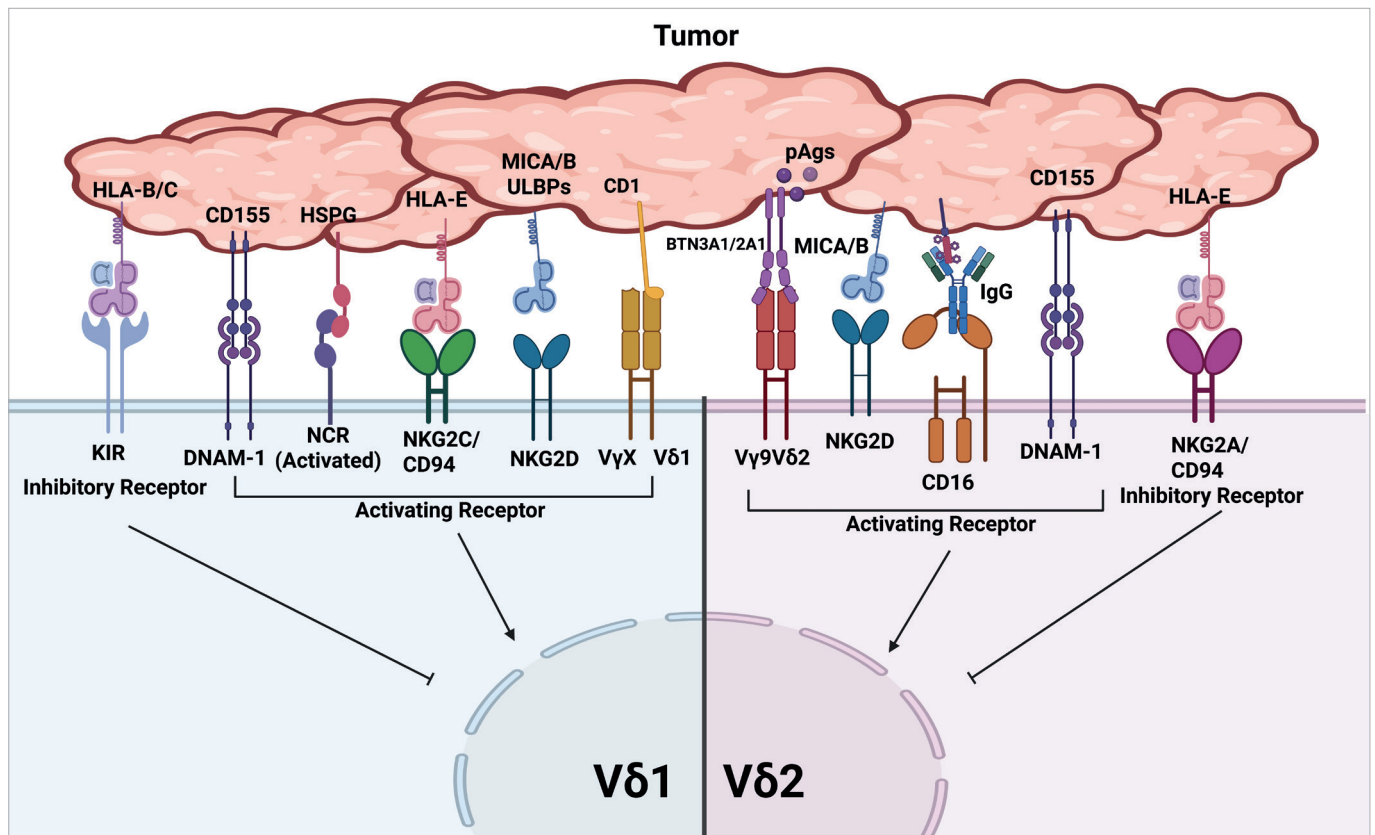


Figure 1. Differential expression of activating and inhibitory NK cell receptors on Vδ1 and Vδ2 $\gamma\delta$ T cells and ligands for NK cell receptors on tumor cells. Created in BioRender. Oh WJ. (2026), <https://BioRender.com/uOfxk8z>.

Activating NKRs expressed on $\gamma\delta$ T cells have differential anti-tumor functions and interact differentially with $\gamma\delta$ -TCRs.

NKG-2D

Recognition of activating ligands is critical to target stressed or cancerous cells by NK or $\gamma\delta$ T cells (19). NKG-2D is the principal activating NKR that recognize several ligands expressed on cancer cells, such as MHC I chain-related molecules A/B (MICA/B), and UL16-binding protein 1–6 in human. NKG-2D exists as a homodimer with a short cytoplasmic tail lacking intrinsic signaling motifs, requiring association with adaptor proteins for signal transduction (20). In humans, NKG-2D exclusively associates with DAP10, forming a hexameric complex consisting of one NKG-2D homodimer and two DAP10 homodimers (21). Ligands for NKG-2D are almost absent on the surface of normal cells but are upregulated in tumor cells, which sensitizes tumor cells to destruction by NKG-2D-expressing NK, CD8⁺ T, or $\gamma\delta$ T cells (22). However, sustained NKG-2D stimulation in advanced cancer tissues tolerizes NK cell activation. Interestingly, the tolerization to chronic NKG-2D signaling also reduce the sensitivity to other activating NKRs (22,23). This raised a view of NKG-2D as a master regulator of immune responses (22). Furthermore, tumor cells frequently release soluble form of NKG-2D ligands via proteolytic-mediated shedding and block NKG-2D-mediated NK cell activation (24). Therefore, the tumoricidal function of NKG-2D is blocked by tolerization of NKG-2D signaling and soluble NKG-2D ligands within the TME in most instances.

Nearly all human $\gamma\delta$ T cells express NKG2D, although its surface density is lower than that on NK cells (25). Human V γ 9V δ 2 T cells—the dominant V δ 2 subset that recognizes phosphoantigens (pAgs) presented by butyrophilin (BTN) 2A1/3A1 (26)—constitutively express both NKG-2D and DAP10, enabling immediate response to stressed or transformed cells expressing NKG-2D ligands without the need for prior activation (27). In addition to mediating direct activation, NKG-2D provides a potent costimulatory signal that synergizes with $\gamma\delta$ -TCR engagement. When tumor cells simultaneously present pAgs and NKG-2D ligands, coordinated signaling markedly amplifies calcium mobilization and cytotoxicity compared with either receptor alone (28). This dual functionality allows V γ 9V δ 2 T cells to operate through flexible and partially redundant recognition pathways: high NKG2D ligand expression can compensate for weak $\gamma\delta$ -TCR stimulation, whereas strong $\gamma\delta$ -TCR signaling can overcome tumor-mediated downregulation of NKG2D ligands, thereby maintaining robust effector responses against heterogeneous tumor targets (29). NKG-2D may function similarly in V δ 1 $\gamma\delta$ T cells. A special situation where $\gamma\delta$ -TCR directly recognize NKG-2D ligands will be addressed under the section of $\gamma\delta$ -TCRs and their ligands.

NCRs

Three NCRs, NCR1 (NKp46), NCR2 (NKp44), and NCR3 (NKp30), have been characterized as activating receptors with immunoglobulin-like extracellular domain(s) on NK cells and $\gamma\delta$ T cell subsets, particularly tissue-resident V δ 1 cells (30). NCRs are associated with cytoplasmic immunoreceptor tyrosine-based activation motif (ITAM)-containing adaptor proteins for signaling. These receptors enhance anti-tumor immunity by recognizing known or unknown stress-induced ligands and tumor-associated molecules, providing an additional layer of target discrimination beyond NKG-2D and $\gamma\delta$ -TCRs (Table 1).

NCR1 (NKp46)

NCR1 recognizes aberrantly glycosylated tumor cell surface molecules, including heparan sulfate proteoglycans and ecto-calreticulin on ER-stressed cells (10,31). NCR1 is prominently expressed on gut-resident intraepithelial V δ 1 cells, where it mediates potent anti-tumor activity against colorectal cancer. NCR1⁺ V δ 1 cells represent the largest $\gamma\delta$ subset in intestinal intraepithelial lymphocytes, and lower frequencies correlate with faster tumor progression and metastasis (9). NCR1 engagement synergizes with $\gamma\delta$ -TCR signals to lower activation thresholds and amplify IFN- γ production and degranulation, functioning as a critical co-stimulatory molecule for tumor recognition (32).

NCR2 (NKp44)

Unlike NCR1, NCR2 exhibits activation-dependent expression, serving as a marker of activated $\gamma\delta$ T cells (33). It recognizes tumor-associated PCNA exposed on cell surfaces (34), PDGF-DD (35), and MLL5 (36). Among $\gamma\delta$ T cell subsets, the NCR2 expression is selectively induced on V δ 1 T cells following prolonged TCR stimulation with IL-15 or IL-2, while V δ 2 T cells fail to express NCRs above background levels (5,11). NCR2⁺NCR3⁺ V δ 1⁺ T cells demonstrate enhanced cytotoxicity against lymphoid leukemias including chronic lymphocytic leukemia, with NCR2 and NCR3 synergistically contributing to tumor cell targeting (11). The ability to selectively expand NCR2⁺ populations through defined cytokine cocktails has important implications for adoptive cell therapy.

NCR3 (NKp30)

NCR3 exhibits the strongest association with anti-tumor activity among NCR family members. NCR3 recognizes B7-H6, a ligand expressed on tumor cells but absent from

normal tissues (37). Upon B7-H6 binding, NCR3 triggers rapid perforin-granzyme release and FasL upregulation, mediating potent cytotoxicity. Peripheral blood NCR3⁺V δ 1⁺ cells efficiently recognize lymphoid leukemia cells (11). NCR3 is further regulated by alternative splicing. While NCR3a and NCR3b promote cytotoxicity, NCR3c functions as a dominant-negative variant that attenuates activation (38). TGF- β -enriched TME enhances the expression of NCR3c (39), which leads to production of immunosuppressive IL-10 and dampens anti-tumor responses (38).

CD16 (Fc γ RIII)

CD16, a low-affinity Fc receptor for IgG that mediates Ab-dependent cellular cytotoxicity (ADCC), is expressed on a variable proportion of $\gamma\delta$ T cells in peripheral blood (40). CD16 functions as an activation receptor that triggers cytotoxic responses independently of TCR engagement. Upon binding to IgG-coated target cells, CD16 initiates signal transduction through associated adaptor molecules containing ITAMs, resulting in degranulation, cytotoxic granule release, and target cell lysis (41). It appears that a mutually exclusive relationship exists between CD16 and CD3 signaling pathways in $\gamma\delta$ T cells as anti-CD3 Abs have been shown to inhibit CD16-mediated ADCC activity. In addition to cytotoxicity, CD16 engagement induces production of pro-inflammatory cytokines such as TNF- α and IFN- γ , which contribute to the overall anti-tumor immune responses (13,41).

The $\gamma\delta$ T cells are heterogeneous concerning the CD16 expression in different donors and $\gamma\delta$ T cell subsets. The proportion of CD16⁺ cells varies considerably among individuals and even within the same individual over time (42). The CD16 expression distinguishes two functionally distinct populations within V δ 2⁺ T cells: CD16⁺V δ 2⁺ T cells primarily respond to pAgs and exhibit TCR-dependent cytotoxicity, whereas CD16⁺V δ 2⁺ T cells demonstrate a pronounced capacity for ADCC (43). This functional dichotomy may suggest that TCR-dependent cytotoxicity and ADCC are separately performed by distinct population and that the $\gamma\delta$ -TCR in CD16⁺V δ 2⁺ T cells have permissive or developmental functions. The CD16-based functional dichotomy also applies to V δ 2⁻ T cells as CD16 is predominantly expressed by V δ 2⁻ $\gamma\delta$ T cells exhibiting a CD45RA⁺CD27⁻ terminally differentiated effector memory phenotype in cytomegalovirus (CMV)-infected individuals (13,44).

The expression of CD16 on $\gamma\delta$ T cells provides a mechanistic rationale for combining adoptive $\gamma\delta$ T cell therapy with therapeutic monoclonal Abs (45). Several studies have demonstrated that CD16-expressing V δ 2 T cells exhibit enhanced cytotoxicity against tumor cells in the presence of clinically approved therapeutic Abs, including rituximab (anti-CD20), trastuzumab (anti-human epidermal growth factor receptor 2), and cetuximab (anti-epidermal growth factor receptor), through ADCC mechanisms (46,47).

Notably, the CD16 expression has been proposed as a biomarker for donor selection in adoptive V δ 2⁺ T cell therapy, as CD16^{high} V δ 2 T cells demonstrate a superior anti-tumor activity both *in vitro* and in preclinical models (48). This approach may help overcome donor-to-donor variability, which has been a significant challenge in $\gamma\delta$ T cell-based immunotherapy.

NKG-2C

Classical MHC class I-a (MHC-Ia; HLA-A, HLA-B, and HLA-C) and non-classical MHC class I-b (MHC-Ib; HLA-E, HLA-F, and HLA-G) are major ligands for inhibitory or activating NKR (49). Although the recognition of MHC-I by inhibitory NKRs is critical to protect host cells, activating NKRs recognizing MHC-I have significant roles in defense against

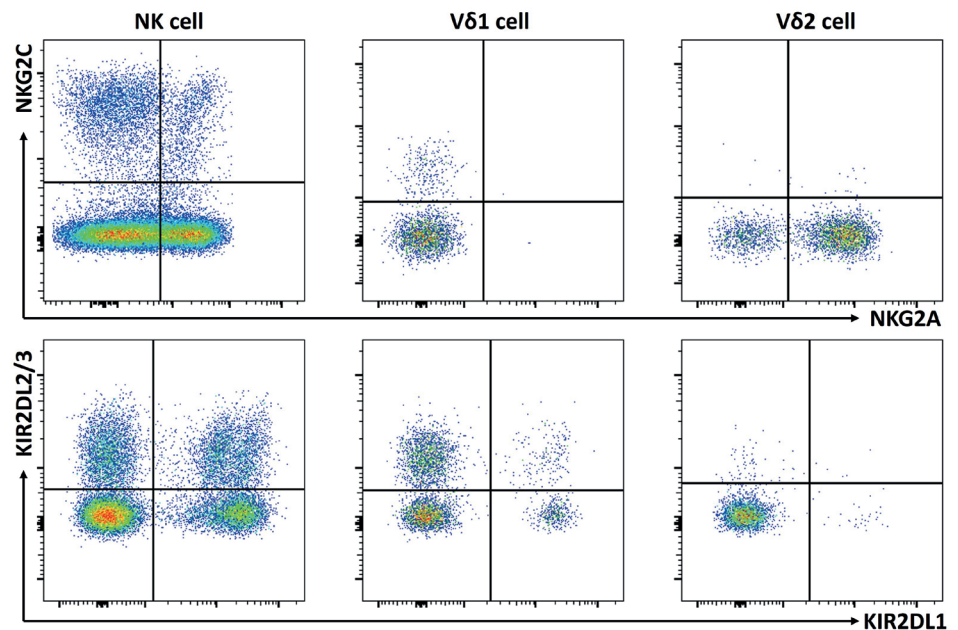


Figure 2. Distinct expression of activating NK cell receptor NKG-2C and inhibitory NK cell receptors KIR2DL1, KIR2DL2/3, and NKG-2A in blood NK cells and blood V δ 1 and V δ 2 $\gamma\delta$ T cells.

infection or cancer cells. NKG-2A and NKG-2C are respective inhibitory and activating C-type lectin-like receptors that forms a heterodimer with CD94 and recognizes the non-classical MHC class I molecule HLA-E (50). Unlike its inhibitory counterpart NKG-2A, CD94/NKG2C associates with the adaptor molecule DNAX activation protein of 12 kDa (DAP12) containing ITAMs, thereby transducing activating signals upon ligand engagement (50,51). The NKG2C expression was shown to be markedly increased on V δ 1 T cells by viral infections, including CMV and HIV (Fig. 2) (52). Notably, the HIV infection induced a profound decrease of V δ 1 T cells bearing NKG-2A along with a drastic expansion of a distinct population of V δ 1 T cells expressing NKG-2C (52). This shift in the NKG-2A/NKG-2C balance on V δ 1 T cells appears to be a critical immunological adaptation in response to viral infection, enhancing the V δ 1 T cell-mediated cytotoxicity against infected target cells (52). These findings indicate that the induction of NKG-2C expression on V δ 1 T cells may play a key role in the destruction of HIV-infected CD4 T cells during disease progression. Similarly, in the context of CMV infection, NKG-2C⁺V δ 1⁺ $\gamma\delta$ T cells are preferentially expanded and exhibit functional characteristics consistent with anti-viral immunity (44,53).

Implication of differential activating NKR on V δ 1 and V δ 2 cells for their clinical utilization

The differences of the activating NKRs in V δ 1 and V δ 2 cells have important implications for $\gamma\delta$ T cell-based immunotherapy. The constitutive NKG-2D expression and pAg responsiveness of V δ 2 T cells make them broadly applicable against NKG-2D ligand-expressing tumors, with the CD16⁺ subset providing additional ADCC capacity for combination with therapeutic Abs. In contrast, V δ 1 T cells exhibit a more diverse repertoire including tissue-specific expression of NCRs and NKG2C, positioning this subset as potent effectors against solid tumors in specific anatomical sites. Understanding these differential receptor usage patterns will be essential for designing subset-targeted therapeutic strategies that harness the unique strengths of each $\gamma\delta$ T cell population.

Next, it is worthy to mention that the activating NKRs can lead to the protumor inflammation and angiogenesis depending on TME. Notably, NKG2D signaling has been shown to regulate IL-17A-producing $\gamma\delta$ T cells in murine tumor models, where blockade of NKG2D reduces IL-17 production and limits tumor progression (54). For other activating NKRs, direct evidence for their pro-tumor function is limited, but their roles in protumor $\gamma\delta$ T cell responses need to be investigated in the future.

TWO CLASSES OF NK INHIBITORY RECEPTORS

Inhibitory NKRs are required to protect host cells while destroying cancerous or infected cells. The requirement of inhibitory NKRs in NK cell development has been suggested as different models of NK cell education such as licensing, disarming, rheostat model, and tuning models (55). $\gamma\delta$ T cells express different combinations of inhibitory NKRs including NKG-2A and KIRs. Differential expression of NKG-2A, NKG-2C, KIR2DL1, and KIR2DL2/3 was shown in NK, V δ 1, and V δ 2 cells (Fig. 2). It is important to know whether the inhibitory NKRs function in $\gamma\delta$ T cells similarly to NK cells. Therefore, we discuss the functional significance of two classes of inhibitory NKRs that recognize the inhibitory ligands in different ways. Here we focus on MHC-I-binding inhibitory receptors, NKG-2A and KIRs.

NKG-2A

NKG-2A is principal inhibitory NKR for NK cell function and development (56). The binding of NKG-2A to HLA-E harboring MHC-I leader peptide transmits an inhibitory signal via its intracytoplasmic immune-receptor tyrosine-based inhibition motifs (ITIMs), recruiting tyrosine phosphatases SHP-1 and SHP-2 (57). This interaction avoids collateral damage of healthy cells during NK cell killing of infected or cancer cells, but cancer cells utilize this interaction to escape from the anti-tumor activities of NKG2A⁺ tumor-infiltrating lymphocytes (58). Therefore, NKG-2A is an immune checkpoint receptor for NK cells and the NKG-2A-HLA-E interaction can be exploited for cancer immunotherapy (59).

Beyond protection of normal cells and immune evasion of cancer cells, NKG-2A has another important function for NK cell education, a process where inhibitory receptor engagement tunes the cell's responsiveness: NK cells that successfully engage with MHC-I or other inhibitory ligands via inhibitory NKRs obtain hyper-responsive effector functions, whereas NK cells lacking the inhibitory receptor engagement become hyporesponsive (56). The licensing hypothesis proposes that NK cells gain the ability to target cells such as tumor cells upon engagement of self MHC-I by inhibitory NKRs during NK cell development (60). Licensed NK cells exhibit enhanced responsiveness upon stimulation through activating NKRs, whereas NK cells lacking cognate self-MHC-I ligands remain hyporesponsive (61). On the other hands, the “disarming” model suggests that NK cells lacking cognate self-MHC-I ligands experience persistent overstimulation, leading to functional anergy and hyporesponsiveness through chronic activation-induced tolerance (62). This model requires sustained low-level stimulation by endogenous activating ligands, resulting in downregulation of effector molecules such as perforin and granzyme and impaired signal transduction (63,64). It is interesting to ask whether these education processes via inhibitory NKRs also applies to $\gamma\delta$ T cells.

The expression of NKG-2A is observed in V δ 2⁺ T cells, but not in V δ 2⁻ T cells (Fig. 2) (16). The educational or licensing effect of NKG-2A that increases anti-tumor effector function

is seen in NKG-2A⁺V δ 2⁺ T cells. However, it is not clear whether NKG-2A⁺V δ 2⁺ T cells can overcome the immune evasion of cancer cells that upregulate of HLA-E. It may be desirable to unlock the immune checkpoint signaling by treatment with anti-NKG-2A Ab when NKG-2A⁺V δ 2⁺ T cells are abundant in the TME (65).

KIRs

KIRs had been regarded as multiple inhibitory NKRs that recognize MHC-I directly and therefore KIRs and NKG-2A are thought to constitute constellation of inhibitory NKRs that perform the immunosurveillance based on the “missing self-hypothesis” of NK cells, where NK cells can only recognize and kill cell lines that lost the expression of MHC-I (66). KIRs are now shown to be a diverse and polymorphic family of glycoproteins that have both inhibitory and activating NKRs recognizing MHC-I and non-MHC-I ligands. Gene contents of different KIR haplotypes are variable with different numbers of inhibitory and activating KIR genes (67). Group A haplotypes have a fixed and limited number of genes and include most inhibitory KIR genes, namely KIR2DL1 (recognizing HLA-C group 2, HLA-C2), KIR2DL3 (recognizing HLA-C group 1, HLA-C1), KIR3DL1 (recognizing HLA-Bw4); KIR3DL2 (recognizing HLA-A3/11), and KIR2DL4 (recognizing HLA-G) (68-71). Group B haplotypes contain a greater gene content and carry more activating KIR genes such as KIR2DS1 (recognizing HLA-C2), KIR2DS2 (recognizing HLA-C1), KIR2DS3 (little surface expression), KIR2DS5 (recognizing HLA-C2), and KIR3DS1 (recognizing HLA-F) than group A haplotypes and also have some inhibitory KIR genes such as KIR2DL1, KIR2DL2 (recognizing HLA-C1), and KIR2DL5A/B (recognizing CD155) (72-77). The KIR haplotype of each individual determines the individually differential activities of NK and $\gamma\delta$ T cells.

Expression of KIRs is seen in V δ 1 cells, but not in V δ 2 cells (Fig. 2). It appears that the regulation of circulating V δ 2 cells is preferentially dependent on NKG-2A whereas tissue-resident V δ 1 cells are regulated more preferentially by KIRs. It would be important to investigate whether V δ 1 cells from group B KIR haplotype have more enhanced anti-tumor activity than those from group A KIR haplotype. It is also worthwhile to note that the activating KIRs from group B haplotypes may counteract against the effects of the inhibitory KIRs. Furthermore, the KIR regulation of V δ 1 $\gamma\delta$ T cells can be influenced by the MHC haplotype of the cancer patients. If cancer patients are homozygous for HLA-C group 1 (C1; asparagine at position 80) or -group 2 (C2; Lysine at position 80), the licensing or inhibition by respective KIR2DL2/3 or KIR2DL1 are not functional (78). This inhibition involves the signaling pathways mediated by their ITIM domains in the cytoplasmic tail of inhibitory KIRs. In summary, the KIR and MHC haplotypes may influence on the anti-cancer immunity by NK and $\gamma\delta$ T cells and even by KIR⁺ $\alpha\beta$ T cells.

$\gamma\delta$ TCRs AND THEIR LIGANDS

pAg recognition by V γ 9V δ 2 T cells

V γ 9V δ 2 T cells are activated by pAgs, which are metabolites of the mevalonate pathway that is enhanced in highly metabolic cancer cells. The most potent pAgs include isopentenyl pyrophosphate (IPP) produced by tumor cells and microbe-derived (E)-4-hydroxy-3-methyl-but-2-enyl pyrophosphate, which is approximately 10,000-fold more potent than IPP (79,80). The recognition of pAgs by V γ 9V δ 2 T cells is mediated through a unique mechanism involving butyrophilin family molecules BTN3A1 and BTN2A1 (26). PAgs bind to the intracellular B30.2 domain of BTN3A1, triggering a conformational change that

facilitates the formation of BTN2A1-BTN3A1 heteromeric complexes on the cell surface for $V\gamma 9V\delta 2$ -TCR engagement (81).

Recent structural studies have revealed that the $V\gamma 9$ and $V\delta 2$ chains of the TCR recognize distinct components of the butyrophilin complex through spatially separated binding sites. BTN2A1 engages the lateral surface of the $V\gamma 9$ chain through germline-encoded regions, independent of CDR3 sequences (26). This interaction is constitutive and provides a platform for TCR engagement. Concurrently, BTN3A1 or the closely related BTN3A2 binds to the apical surface of the TCR, engaging the $V\delta 2$ chain along with germline-encoded regions and CDR3 motifs of both chains (26,81). This dual recognition mechanism, whereby the γ and δ chains independently engage distinct butyrophilin molecules, creates a “composite ligand” model for pAg sensing. The spatial separation of these binding sites allows the TCR to simultaneously survey both BTN2A1 and pAg-activated BTN3A1, thereby translating intracellular pAg accumulation into T cell activation (26,81).

Heterogeneity of $V\gamma 9V\delta 2$ T cell repertoires: $J\gamma 2$ vs. $J\gamma P$

Although $V\gamma 9V\delta 2$ T cells are conventionally regarded as a homogeneous population, recent repertoire analyses have revealed significant heterogeneity in their J segment usage. Canonical $V\gamma 9V\delta 2$ T cells in peripheral blood predominantly utilize $V\gamma 9J\gamma P$ sequences that pair with the $C\gamma 1$ constant segment, whereas tumor-infiltrating $V\gamma 9V\delta 2$ T cells from glioblastoma patients are predominated by $V\gamma 9J\gamma 2$ sequences that utilize the $C\gamma 2$ segment (82). Importantly, $V\gamma 9J\gamma 2$ - $V\delta 2$ T cells exhibit shorter CDR3 γ sequences and some clonotypes have distinct structural features that may confer alternative antigenic specificities relevant for tumor immunity (82).

Ligand recognition by $V\delta 1$ T cells

In contrast to $V\gamma 9V\delta 2$ T cells, $V\delta 1$ T cells exhibit broader and more diverse ligand recognition patterns. $\gamma\delta$ -TCR recognition of Ags is not restricted by MHC and similar to the interaction between Ab and Ag. Additionally, some $V\delta 1$ T cells can recognize lipid Ags presented by CD1 molecules, as well as other stress-induced ligands that are upregulated on transformed or infected cells (83,84). This diverse ligand recognition repertoire positions $V\delta 1$ T cells as important sentinels in tissue surveillance and anti-tumor immunity.

Interestingly, $V\delta 1$ $\gamma\delta$ -TCRs specific to MICA and MICB have been reported, which bind MICA with a lower affinity than NKG-2D (85,86). The interaction between $V\delta 1$ TCR and MICA has been structurally characterized, revealing that both the TCR and the NK activating receptor NKG2D can independently bind to MICA, providing dual recognition mechanisms (85). Despite a lower affinity for MICA, the $V\delta 1$ $\gamma\delta$ -TCR binding to MICA was unusually stable and competed with the binding between NKG-2D and MICA. These MICA/B-binding $V\delta 1$ $\gamma\delta$ T cells may be important players that perform anti-cancer cytotoxicity in the NKG-2D-tolerizing TME. This dual recognition system provides $V\delta 1$ T cells with enhanced discrimination, combining the rapid stress-sensing capacity of NKG-2D with TCR-mediated specificity that prevents inappropriate activation against transiently stressed healthy cells. The functional outcome is a highly selective recognition system that responds to pathological cells while minimizing collateral damage to healthy tissues (87).

In summary, $\gamma\delta$ T cells employ diverse Ag recognition strategies that distinguish them from conventional $\alpha\beta$ T cells and $V\delta 1$ and $V\delta 2$ $\gamma\delta$ T cells show differential strategies to detect abnormal cells.

INTEGRATIVE MODELS OF $\gamma\delta$ -TCR AND NKR SIGNAL COORDINATION

Rather than functioning as parallel or redundant pathways, $\gamma\delta$ -TCRs and NKRs form an integrated signaling network that enables $\gamma\delta$ T cells to flexibly adapt to heterogeneous TMEs (88). In settings characterized by high expression of tumor-associated stress ligands, activating NKRs can dominate signal integration and drive rapid cytotoxic responses largely independent of $\gamma\delta$ -TCR engagement (11,89). In contrast, in tumors with low or heterogeneous stress-ligand expression, $\gamma\delta$ -TCR signaling may provide the primary activation input, with NKRs functioning predominantly as co-stimulatory or inhibitory modulators that fine-tune effector responses.

Within the TME, sustained exposure to abundant soluble or membrane-bound NKG2D ligands can result in chronic NKG2D engagement and functional exhaustion, thereby limiting durable anti-tumor activity (23,90). Under these conditions, alternative activating NKRs, such as NCRs, and/or specific $\gamma\delta$ -TCR clonotypes may become increasingly important for maintaining effector function (9,30,91). Notably, V δ 1 $\gamma\delta$ T cells, which exhibit broader TCR diversity and distinct tissue localization, may play a key role in sustaining anti-tumor immunity in such contexts (89,92). By contrast, semi-invariant V δ 2 $\gamma\delta$ -TCRs recognize shared pAg in the context of BTN3A1/BTN2A1; however, their functional outcomes can diverge markedly (26,93). Depending on the balance between activating and inhibitory NKR signaling within the TME, V δ 2 $\gamma\delta$ T cells may contribute to either anti-tumor cytotoxicity or immunoregulatory, potentially pro-tumor, responses. Collectively, $\gamma\delta$ -TCRs endow $\gamma\delta$ T cells with a broad and adaptable repertoire of Ag recognition that complements NKR-mediated sensing, distinguishing $\gamma\delta$ T cells from NK cells.

CONCLUDING REMARK

Conventional CD8⁺ CTLs are widely regarded as the principal effector population responsible for eliminating malignant cells; however, many tumors develop robust mechanisms to evade CTL-mediated immunity. A major vulnerability of CTLs lies in their strict dependence on peptide–MHC-I presentation for Ag recognition (94). The peptide–MHC-I processing and presentation machinery includes multiple genes—such as *B2M*, *HLA-A/B/C*, *TAP1/2*, *ERAP1/2*, *NLRC5*, and components of the interferon–JAK/STAT axis—that are frequently mutated, epigenetically silenced, or transcriptionally downregulated across diverse cancer types, resulting in stable and heritable loss of MHC-I expression and subsequent immune escape (95,96). Moreover, unlike B cells, T cells lack AID-mediated somatic hypermutation and affinity maturation; thus, the breadth and specificity of CTL responses are constrained by the pre-existing TCR repertoire and its limited ability to adapt to rapidly evolving tumor neoantigens. These limitations highlight that despite their numerical predominance within the TME, CTLs can become functionally restricted under conditions of tumor immunoediting or antigenic drift. This has intensified interest in MHC-unrestricted CTLs which can recognize ligands independently of classical MHC-I. Among these, $\gamma\delta$ T cells have emerged as especially promising therapeutic candidates because their cytotoxicity can be readily enhanced through stimulation of afore-discussed activating NKRs or blockade of inhibitory NKRs (97,98). Therefore, it is worthwhile to utilize double-faced features of $\gamma\delta$ T cells as innate NK cells and adaptive T cells in appropriate contexts. Consequently, both autologous and allogeneic $\gamma\delta$ T-cell-based immunotherapies, including ex vivo expanded

V γ 9V δ 2 T cells, CAR- $\gamma\delta$ platforms, and pAg-sensitization strategies, are now being actively evaluated in pre-clinical models and clinical trials (99). Collectively, these features position $\gamma\delta$ T cells as a compelling next-generation cytotoxic platform capable of complementing or surpassing conventional CTL-based cancer immunotherapies, particularly in tumors that have evolved to escape adaptive immune pressure.

ACKNOWLEDGEMENTS

This work was supported by a National Research Foundation of Korea grant funded by the Korea government (MSIT) (RS-2024-00405650).

REFERENCES

1. Lee HW, Chung YS, Kim TJ. Heterogeneity of human $\gamma\delta$ T cells and their role in cancer immunity. *Immune Netw* 2020;20:e5. [PUBMED](#) | [CROSSREF](#)
2. You H, Zhang X, Chen H, Liu C, Teng D, Han J, Chen M, Pang Y, Zhang J, Cai M, et al. $\gamma\delta$ T-cell autoresponses to ectopic membrane proteins: a new type of pattern recognition. *Cell Mol Immunol* 2025;22:356-370. [PUBMED](#) | [CROSSREF](#)
3. Silva-Santos B, Mensurado S, Coffelt SB. $\gamma\delta$ T cells: pleiotropic immune effectors with therapeutic potential in cancer. *Nat Rev Cancer* 2019;19:392-404. [PUBMED](#) | [CROSSREF](#)
4. Melandri D, Zlatareva I, Chaleil RAG, Dart RJ, Chancellor A, Nussbaumer O, Polyakova O, Roberts NA, Wesch D, Kabelitz D, et al. The $\gamma\delta$ TCR combines innate immunity with adaptive immunity by utilizing spatially distinct regions for agonist selection and antigen responsiveness. *Nat Immunol* 2018;19:1352-1365. [PUBMED](#) | [CROSSREF](#)
5. Silva-Santos B, Strid J. Working in "NK Mode": natural killer group 2 member D and natural cytotoxicity receptors in stress-surveillance by $\gamma\delta$ T cells. *Front Immunol* 2018;9:851. [PUBMED](#) | [CROSSREF](#)
6. Fleming C, Morrissey S, Cai Y, Yan J. $\gamma\delta$ T cells: unexpected regulators of cancer development and progression. *Trends Cancer* 2017;3:561-570. [PUBMED](#) | [CROSSREF](#)
7. Sanz M, Mann BT, Ryan PL, Bosque A, Pennington DJ, Hackstein H, Soriano-Sarabia N. Deep characterization of human $\gamma\delta$ T cell subsets defines shared and lineage-specific traits. *Front Immunol* 2023;14:1148988. [PUBMED](#) | [CROSSREF](#)
8. Kim M, Kim H, Han M, Hwang HJ, Kim H, Im HJ, Kim N, Koh KN. Characteristics of human peripheral blood $\gamma\delta$ T cells expanded with zoledronate. *Anticancer Res* 2021;41:6031-6038. [PUBMED](#) | [CROSSREF](#)
9. Mikulak J, Oriolo F, Bruni E, Roberto A, Colombo FS, Villa A, Bosticardo M, Bortolomai I, Lo Presti E, Meraviglia S, et al. NKp46-expressing human gut-resident intraepithelial V δ 1 T cell subpopulation exhibits high antitumor activity against colorectal cancer. *JCI Insight* 2019;4:e125884. [PUBMED](#) | [CROSSREF](#)
10. Mandelboim O, Lieberman N, Lev M, Paul L, Arnon TI, Bushkin Y, Davis DM, Strominger JL, Yewdell JW, Porgador A. Recognition of haemagglutinins on virus-infected cells by NKp46 activates lysis by human NK cells. *Nature* 2001;409:1055-1060. [PUBMED](#) | [CROSSREF](#)
11. Correia DV, Fogli M, Hudspeth K, da Silva MG, Mavilio D, Silva-Santos B. Differentiation of human peripheral blood V δ 1+ T cells expressing the natural cytotoxicity receptor NKp30 for recognition of lymphoid leukemia cells. *Blood* 2011;118:992-1001. [PUBMED](#) | [CROSSREF](#)
12. Pende D, Parolini S, Pessino A, Sivori S, Augugliaro R, Morelli L, Marcenaro E, Accame L, Malaspina A, Biassoni R, et al. Identification and molecular characterization of NKp30, a novel triggering receptor involved in natural cytotoxicity mediated by human natural killer cells. *J Exp Med* 1999;190:1505-1516. [PUBMED](#) | [CROSSREF](#)
13. Couzi L, Pitard V, Sicard X, Garrigue I, Hawchar O, Merville P, Moreau JF, Déchanet-Merville J. Antibody-dependent anti-cytomegalovirus activity of human $\gamma\delta$ T cells expressing CD16 (Fc γ RIIIa). *Blood* 2012;119:1418-1427. [PUBMED](#) | [CROSSREF](#)
14. López-Botet M, Muntasell A, Vilches C. The CD94/NKG2C+ NK-cell subset on the edge of innate and adaptive immunity to human cytomegalovirus infection. *Semin Immunol* 2014;26:145-151. [PUBMED](#) | [CROSSREF](#)

15. Pitard V, Roumanes D, Lafarge X, Couzi L, Garrigue I, Lafon ME, Merville P, Moreau JF, Déchanet-Merville J. Long-term expansion of effector/memory Vdelta2-gammadelta T cells is a specific blood signature of CMV infection. *Blood* 2008;112:1317-1324. [PUBMED](#) | [CROSSREF](#)
16. Cazzetta V, Bruni E, Terzoli S, Carena C, Franzese S, Piazza R, Marzano P, Donadon M, Torzilli G, Cimino M, et al. NKG2A expression identifies a subset of human Vdelta2 T cells exerting the highest antitumor effector functions. *Cell Reports* 2021;37:109871. [PUBMED](#) | [CROSSREF](#)
17. Couzi L, Pitard V, Netzer S, Garrigue I, Lafon ME, Moreau JF, Taupin JL, Merville P, Déchanet-Merville J. Common features of gammadelta T cells and CD8(+) alphabeta T cells responding to human cytomegalovirus infection in kidney transplant recipients. *J Infect Dis* 2009;200:1415-1424. [PUBMED](#) | [CROSSREF](#)
18. Palmer WH, Leaton LA, Campos Codo A, Crute B, Roest J, Zhu S, Petersen J, Tobin RP, Hume PS, Stone M, et al. Polymorphic KIR3DL3 expression modulates tissue-resident and innate-like T cells. *Sci Immunol* 2023;8:eade5343. [PUBMED](#) | [CROSSREF](#)
19. Lanier LL. NKG2D receptor and its ligands in host defense. *Cancer Immunol Res* 2015;3:575-582. [PUBMED](#) | [CROSSREF](#)
20. Wu J, Song Y, Bakker AB, Bauer S, Spies T, Lanier LL, Phillips JH. An activating immunoreceptor complex formed by NKG2D and DAP10. *Science* 1999;285:730-732. [PUBMED](#) | [CROSSREF](#)
21. Garrity D, Call ME, Feng J, Wucherpfennig KW. The activating NKG2D receptor assembles in the membrane with two signaling dimers into a hexameric structure. *Proc Natl Acad Sci U S A* 2005;102:7641-7646. [PUBMED](#) | [CROSSREF](#)
22. Wensveen FM, Jelenčić V, Polić B. NKG2D: a master regulator of immune cell responsiveness. *Front Immunol* 2018;9:441. [PUBMED](#) | [CROSSREF](#)
23. Coudert JD, Scarpellino L, Gros F, Vivier E, Held W. Sustained NKG2D engagement induces cross-tolerance of multiple distinct NK cell activation pathways. *Blood* 2008;111:3571-3578. [PUBMED](#) | [CROSSREF](#)
24. Zingoni A, Vulpis E, Loconte L, Santoni A. NKG2D ligand shedding in response to stress: role of ADAM10. *Front Immunol* 2020;11:447. [PUBMED](#) | [CROSSREF](#)
25. Dong R, Zhang Y, Xiao H, Zeng X. Engineering $\gamma\delta$ T cells: recognizing and activating on their own way. *Front Immunol* 2022;13:889051. [PUBMED](#) | [CROSSREF](#)
26. Zhang M, Wang Y, Cai N, Qu Y, Ma X, Xue J, Chen X, Zhang X, Xiao J, Zhang Y. Structures of butyrophilin multimers reveal a plier-like mechanism for Vgamma9Vdelta2 T cell receptor activation. *Immunity* 2025;58:1660-1669.e7. [PUBMED](#) | [CROSSREF](#)
27. Rincon-Orozco B, Kunzmann V, Wrobel P, Kabelitz D, Steinle A, Herrmann T. Activation of Vgamma9Vdelta2 T cells by NKG2D. *J Immunol* 2005;175:2144-2151. [PUBMED](#) | [CROSSREF](#)
28. Nedellec S, Sabourin C, Bonneville M, Scotet E. NKG2D costimulates human Vgamma9Vdelta2 T cell antitumor cytotoxicity through protein kinase C theta-dependent modulation of early TCR-induced calcium and transduction signals. *J Immunol* 2010;185:55-63. [PUBMED](#) | [CROSSREF](#)
29. Chauvin C, Joalland N, Perroteau J, Jarry U, Lafrance L, Willem C, Retière C, Oliver L, Gratas C, Gautreau-Rolland L, et al. NKG2D controls natural reactivity of Vgamma9Vdelta2 T lymphocytes against mesenchymal glioblastoma cells. *Clin Cancer Res* 2019;25:7218-7228. [PUBMED](#) | [CROSSREF](#)
30. Barrow AD, Martin CJ, Colonna M. The natural cytotoxicity receptors in health and disease. *Front Immunol* 2019;10:909. [PUBMED](#) | [CROSSREF](#)
31. Kruse PH, Matta J, Ugolini S, Vivier E. Natural cytotoxicity receptors and their ligands. *Immunol Cell Biol* 2014;92:221-229. [PUBMED](#) | [CROSSREF](#)
32. Biassoni R, Cantoni C, Pende D, Sivori S, Parolini S, Vitale M, Bottino C, Moretta A. Human natural killer cell receptors and co-receptors. *Immunol Rev* 2001;181:203-214. [PUBMED](#) | [CROSSREF](#)
33. von Lilienfeld-Toal M, Nattermann J, Feldmann G, Sievers E, Frank S, Strehl J, Schmidt-Wolf IG. Activated gammadelta T cells express the natural cytotoxicity receptor natural killer p 44 and show cytotoxic activity against myeloma cells. *Clin Exp Immunol* 2006;144:528-533. [PUBMED](#) | [CROSSREF](#)
34. Rosental B, Brusilovsky M, Hadad U, Oz D, Appel MY, Afergan F, Yossef R, Rosenberg LA, Aharoni A, Cerwenka A, et al. Proliferating cell nuclear antigen is a novel inhibitory ligand for the natural cytotoxicity receptor NKp44. *J Immunol* 2011;187:5693-5702. [PUBMED](#) | [CROSSREF](#)
35. Barrow AD, Edeling MA, Trifonov V, Luo J, Goyal P, Bohl B, Bando JK, Kim AH, Walker J, Andahazy M, et al. Natural killer cells control tumor growth by sensing a growth factor. *Cell* 2018;172:534-548.e19. [PUBMED](#) | [CROSSREF](#)
36. Baychelier F, Sennepin A, Ermonval M, Dorgham K, Debré P, Vieillard V. Identification of a cellular ligand for the natural cytotoxicity receptor NKp44. *Blood* 2013;122:2935-2942. [PUBMED](#) | [CROSSREF](#)

37. Brandt CS, Baratin M, Yi EC, Kennedy J, Gao Z, Fox B, Haldeman B, Ostrander CD, Kaifu T, Chabannon C, et al. The B7 family member B7-H6 is a tumor cell ligand for the activating natural killer cell receptor NKp30 in humans. *J Exp Med* 2009;206:1495-1503. [PUBMED](#) | [CROSSREF](#)
38. Delahaye NF, Rusakiewicz S, Martins I, Ménard C, Roux S, Lyonnet L, Paul P, Sarabi M, Chaput N, Semeraro M, et al. Alternatively spliced NKp30 isoforms affect the prognosis of gastrointestinal stromal tumors. *Nat Med* 2011;17:700-707. [PUBMED](#) | [CROSSREF](#)
39. Siewiera J, Gouilly J, Hocine HR, Cartron G, Levy C, Al-Daccak R, Jabrane-Ferrat N. Natural cytotoxicity receptor splice variants orchestrate the distinct functions of human natural killer cell subtypes. *Nat Commun* 2015;6:10183. [PUBMED](#) | [CROSSREF](#)
40. Capuano C, Pighi C, Battella S, De Federicis D, Galandrini R, Palmieri G. Harnessing CD16-mediated NK cell functions to enhance therapeutic efficacy of tumor-targeting mAbs. *Cancers (Basel)* 2021;13:2500. [PUBMED](#) | [CROSSREF](#)
41. Lafont V, Liautard J, Liautard JP, Favero J. Production of TNF- α by human V γ 9V δ 2 T cells via engagement of Fc γ RIIIA, the low affinity type 3 receptor for the Fc portion of IgG, expressed upon TCR activation by nonpeptidic antigen. *J Immunol* 2001;166:7190-7199. [PUBMED](#) | [CROSSREF](#)
42. Bodman-Smith MD, Anand A, Durand V, Youinou PY, Lydyard PM. Decreased expression of Fc γ RIII (CD16) by $\gamma\delta$ T cells in patients with rheumatoid arthritis. *Immunology* 2000;99:498-503. [PUBMED](#) | [CROSSREF](#)
43. He X, Liang H, Hong K, Li H, Peng H, Zhao Y, Jia M, Ruan Y, Shao Y. The potential role of CD16+ V γ 2V δ 2 T cell-mediated antibody-dependent cell-mediated cytotoxicity in control of HIV type 1 disease. *AIDS Res Hum Retroviruses* 2013;29:1562-1570. [PUBMED](#) | [CROSSREF](#)
44. Knight A, Madrigal AJ, Grace S, Sivakumaran J, Kottaridis P, Mackinnon S, Travers PJ, Lowdell MW. The role of V δ 2-negative $\gamma\delta$ T cells during cytomegalovirus reactivation in recipients of allogeneic stem cell transplantation. *Blood* 2010;116:2164-2172. [PUBMED](#) | [CROSSREF](#)
45. Shin MH, Oh E, Minn D. Current developments in NK cell engagers for cancer immunotherapy: focus on CD16A and NKp46. *Immune Netw* 2024;24:e34. [PUBMED](#) | [CROSSREF](#)
46. Tokuyama H, Hagi T, Mattarollo SR, Morley J, Wang Q, So HF, Moriyasu F, Nieda M, Nicol AJ. V γ 9V δ 2 T cell cytotoxicity against tumor cells is enhanced by monoclonal antibody drugs—rituximab and trastuzumab. *Int J Cancer* 2008;122:2526-2534. [PUBMED](#) | [CROSSREF](#)
47. Kabelitz D, Kalyan S, Oberg HH, Wesch D. Human V δ 2 versus non-V δ 2 $\gamma\delta$ T cells in antitumor immunity. *Onc Immunology* 2013;2:e23304. [PUBMED](#) | [CROSSREF](#)
48. Lee D, Dunn ZS, Guo W, Rosenthal CJ, Penn NE, Yu Y, Zhou K, Li Z, Ma F, Li M, et al. Unlocking the potential of allogeneic V δ 2 T cells for ovarian cancer therapy through CD16 biomarker selection and CAR/IL-15 engineering. *Nat Commun* 2023;14:6942. [PUBMED](#) | [CROSSREF](#)
49. Guethlein LA, Norman PJ, Hilton HG, Parham P. Co-evolution of MHC class I and variable NK cell receptors in placental mammals. *Immunol Rev* 2015;267:259-282. [PUBMED](#) | [CROSSREF](#)
50. Braud VM, Allan DS, O'Callaghan CA, Söderström K, D'Andrea A, Ogg GS, Lazetic S, Young NT, Bell JI, Phillips JH, et al. HLA-E binds to natural killer cell receptors CD94/NKG2A, B and C. *Nature* 1998;391:795-799. [PUBMED](#) | [CROSSREF](#)
51. Lanier LL, Corliss B, Wu J, Phillips JH. Association of DAP12 with activating CD94/NKG2C NK cell receptors. *Immunity* 1998;8:693-701. [PUBMED](#) | [CROSSREF](#)
52. Fausther-Bovendo H, Wauquier N, Cherfils-Vicini J, Cremer I, Debré P, Vieillard V. NKG2C is a major triggering receptor involved in the V δ 1 T cell-mediated cytotoxicity against HIV-infected CD4 T cells. *AIDS* 2008;22:217-226. [PUBMED](#) | [CROSSREF](#)
53. Stankovic S, Davey MS, Shaw EM, Cristiano Y, Snell GI, Westall GP, Brooks AG, Sullivan LC. A potential role for NKG2C+ $\gamma\delta$ T cells as a cellular therapy in combating post-lung transplant cytomegalovirus infection. *J Heart Lung Transplant* 2020;39:S82.
54. Curio S, Edwards SC, Suzuki T, McGovern J, Triulzi C, Yoshida N, Jonsson G, Glauner T, Rami D, Wiesheu R, et al. NKG2D signaling regulates IL-17A-producing $\gamma\delta$ T cells in mice to promote cancer progression. *Discov Immunol* 2022;1:kyac002. [PUBMED](#) | [CROSSREF](#)
55. Boudreau JE, Hsu KC. Natural killer cell education and the response to infection and cancer therapy: stay tuned. *Trends Immunol* 2018;39:222-239. [PUBMED](#) | [CROSSREF](#)
56. He Y, Tian Z. NK cell education via nonclassical MHC and non-MHC ligands. *Cell Mol Immunol* 2017;14:321-330. [PUBMED](#) | [CROSSREF](#)
57. Joncker NT, Fernandez NC, Treiner E, Vivier E, Raulet DH. NK cell responsiveness is tuned commensurate with the number of inhibitory receptors for self-MHC class I: the rheostat model. *J Immunol* 2009;182:4572-4580. [PUBMED](#) | [CROSSREF](#)

58. Abd Hamid M, Wang RZ, Yao X, Fan P, Li X, Chang XM, Feng Y, Jones S, Maldonado-Perez D, Waugh C, et al. Enriched HLA-E and CD94/NKG2A interaction limits antitumor CD8⁺ tumor-infiltrating T lymphocyte responses. *Cancer Immunol Res* 2019;7:1293-1306. [PUBMED](#) | [CROSSREF](#)
59. Li Y, Li Z, Tang Y, Zhuang X, Feng W, Boor PPC, Buschow S, Sprengers D, Zhou G. Unlocking the therapeutic potential of the NKG2A-HLA-E immune checkpoint pathway in T cells and NK cells for cancer immunotherapy. *J Immunother Cancer* 2024;12:e009934. [PUBMED](#) | [CROSSREF](#)
60. Tu MM, Mahmoud AB, Makriganis AP. Licensed and unlicensed NK cells: differential roles in cancer and viral control. *Front Immunol* 2016;7:166. [PUBMED](#) | [CROSSREF](#)
61. Anfossi N, André P, Guia S, Falk CS, Roetynck S, Stewart CA, Bresó V, Frassati C, Reviron D, Middleton D, et al. Human NK cell education by inhibitory receptors for MHC class I. *Immunity* 2006;25:331-342. [PUBMED](#) | [CROSSREF](#)
62. Raulet DH, Vance RE. Self-tolerance of natural killer cells. *Nat Rev Immunol* 2006;6:520-531. [PUBMED](#) | [CROSSREF](#)
63. Tripathy SK, Keyel PA, Yang L, Pingel JT, Cheng TP, Schneeberger A, Yokoyama WM. Continuous engagement of a self-specific activation receptor induces NK cell tolerance. *J Exp Med* 2008;205:1829-1841. [PUBMED](#) | [CROSSREF](#)
64. Fauriat C, Long EO, Ljunggren HG, Bryceson YT. Regulation of human NK-cell cytokine and chemokine production by target cell recognition. *Blood* 2010;115:2167-2176. [PUBMED](#) | [CROSSREF](#)
65. André P, Denis C, Soulas C, Bourbon-Caillet C, Lopez J, Arnoux T, Bléry M, Bonnafous C, Gauthier L, Morel A, et al. Anti-NKG2A mAb is a checkpoint inhibitor that promotes anti-tumor immunity by unleashing both T and NK cells. *Cell* 2018;175:1731-1743.e13. [PUBMED](#) | [CROSSREF](#)
66. Parham P. MHC class I molecules and KIRs in human history, health and survival. *Nat Rev Immunol* 2005;5:201-214. [PUBMED](#) | [CROSSREF](#)
67. Yawata M, Yawata N, Draghi M, Little AM, Partheniou F, Parham P. Roles for HLA and KIR polymorphisms in natural killer cell repertoire selection and modulation of effector function. *J Exp Med* 2006;203:633-645. [PUBMED](#) | [CROSSREF](#)
68. Sim MJ, Stowell J, Sergeant R, Altmann DM, Long EO, Boyton RJ. KIR2DL3 and KIR2DL1 show similar impact on licensing of human NK cells. *Eur J Immunol* 2016;46:185-191. [PUBMED](#) | [CROSSREF](#)
69. Gardiner CM, Guethlein LA, Shilling HG, Pando M, Carr WH, Rajalingam R, Vilches C, Parham P. Different NK cell surface phenotypes defined by the DX9 antibody are due to KIR3DL1 gene polymorphism. *J Immunol* 2001;166:2992-3001. [PUBMED](#) | [CROSSREF](#)
70. Hansasuta P, Dong T, Thananchai H, Weekes M, Willberg C, Aldemir H, Rowland-Jones S, Braud VM. Recognition of HLA-A3 and HLA-A11 by KIR3DL2 is peptide-specific. *Eur J Immunol* 2004;34:1673-1679. [PUBMED](#) | [CROSSREF](#)
71. Faure M, Long EO. KIR2DL4 (CD158d), an NK cell-activating receptor with inhibitory potential. *J Immunol* 2002;168:6208-6214. [PUBMED](#) | [CROSSREF](#)
72. Ren X, Peng M, Xing P, Wei Y, Galbo PM Jr, Corrigan D, Wang H, Su Y, Dong X, Sun Q, et al. Blockade of the immunosuppressive KIR2DL5/PVR pathway elicits potent human NK cell-mediated antitumor immunity. *J Clin Invest* 2022;132:e163620. [PUBMED](#) | [CROSSREF](#)
73. Xiong S, Sharkey AM, Kennedy PR, Gardner L, Farrell LE, Chazara O, Bauer J, Hiby SE, Colucci F, Moffett A. Maternal uterine NK cell-activating receptor KIR2DS1 enhances placentation. *J Clin Invest* 2013;123:4264-4272. [PUBMED](#) | [CROSSREF](#)
74. David G, Djaoud Z, Willem C, Legrand N, Rettman P, Gagne K, Cesbron A, Retière C. Large spectrum of HLA-C recognition by killer Ig-like receptor (KIR)2DL2 and KIR2DL3 and restricted C1 SPECIFICITY of KIR2DS2: dominant impact of KIR2DL2/KIR2DS2 on KIR2D NK cell repertoire formation. *J Immunol* 2013;191:4778-4788. [PUBMED](#) | [CROSSREF](#)
75. VandenBussche CJ, Mulrooney TJ, Frazier WR, Dakshanamurthy S, Hurley CK. Dramatically reduced surface expression of NK cell receptor KIR2DS3 is attributed to multiple residues throughout the molecule. *Genes Immun* 2009;10:162-173. [PUBMED](#) | [CROSSREF](#)
76. Blokhuis JH, Hilton HG, Guethlein LA, Norman PJ, Nemat-Gorgani N, Nakimuli A, Chazara O, Moffett A, Parham P. KIR2DS5 allotypes that recognize the C2 epitope of HLA-C are common among Africans and absent from Europeans. *Immun Inflamm Dis* 2017;5:461-468. [PUBMED](#) | [CROSSREF](#)
77. Garcia-Beltran WF, Hölzemer A, Martrus G, Chung AW, Pacheco Y, Simoneau CR, Rucevic M, Lamothe-Molina PA, Pertel T, Kim TE, et al. Open conformers of HLA-F are high-affinity ligands of the activating NK-cell receptor KIR3DS1. *Nat Immunol* 2016;17:1067-1074. [PUBMED](#) | [CROSSREF](#)
78. Winter CC, Long EO. A single amino acid in the p58 killer cell inhibitory receptor controls the ability of natural killer cells to discriminate between the two groups of HLA-C allotypes. *J Immunol* 1997;158:4026-4028. [PUBMED](#) | [CROSSREF](#)

79. Tanaka Y, Morita CT, Tanaka Y, Nieves E, Brenner MB, Bloom BR. Natural and synthetic non-peptide antigens recognized by human $\gamma\delta$ T cells. *Nature* 1995;375:155-158. [PUBMED](#) | [CROSSREF](#)
80. Eberl M, Hintz M, Reichenberg A, Kollas AK, Wiesner J, Jomaa H. Microbial isoprenoid biosynthesis and human $\gamma\delta$ T cell activation. *FEBS Lett* 2003;544:4-10. [PUBMED](#) | [CROSSREF](#)
81. Fulford TS, Soliman C, Castle RG, Rigau M, Ruan Z, Dolezal O, Seneviratna R, Brown HG, Hanssen E, Hammet A, et al. V γ 9V δ 2 T cells recognize butyrophilin 2A1 and 3A1 heteromers. *Nat Immunol* 2024;25:1355-1366. [PUBMED](#) | [CROSSREF](#)
82. Lee M, Park C, Woo J, Kim J, Kho I, Nam DH, Park WY, Kim YS, Kong DS, Lee HW, et al. Preferential infiltration of unique V γ 9 γ 2-V δ 2 T cells into glioblastoma multiforme. *Front Immunol* 2019;10:555. [PUBMED](#) | [CROSSREF](#)
83. Uldrich AP, Le Nours J, Pellicci DG, Gherardin NA, McPherson KG, Lim RT, Patel O, Beddoe T, Gras S, Rossjohn J, et al. CD1d-lipid antigen recognition by the $\gamma\delta$ TCR. *Nat Immunol* 2013;14:1137-1145. [PUBMED](#) | [CROSSREF](#)
84. Willcox CR, Pitard V, Netzer S, Couzi L, Salim M, Silberzahn T, Moreau JF, Hayday AC, Willcox BE, Déchanet-Merville J. Cytomegalovirus and tumor stress surveillance by binding of a human $\gamma\delta$ T cell antigen receptor to endothelial protein C receptor. *Nat Immunol* 2012;13:872-879. [PUBMED](#) | [CROSSREF](#)
85. Xu B, Pizarro JC, Holmes MA, McBeth C, Groh V, Spies T, Strong RK. Crystal structure of a gammadelta T-cell receptor specific for the human MHC class I homolog MICA. *Proc Natl Acad Sci U S A* 2011;108:2414-2419. [PUBMED](#) | [CROSSREF](#)
86. Groh V, Rhinehart R, Secrist H, Bauer S, Grabstein KH, Spies T. Broad tumor-associated expression and recognition by tumor-derived $\gamma\delta$ T cells of MICA and MICB. *Proc Natl Acad Sci U S A* 1999;96:6879-6884. [PUBMED](#) | [CROSSREF](#)
87. Nielsen MM, Witherden DA, Havran WL. $\gamma\delta$ T cells in homeostasis and host defence of epithelial barrier tissues. *Nat Rev Immunol* 2017;17:733-745. [PUBMED](#) | [CROSSREF](#)
88. Kabelitz D, Serrano R, Kouakanou L, Peters C, Kalyan S. Cancer immunotherapy with $\gamma\delta$ T cells: many paths ahead of us. *Cell Mol Immunol* 2020;17:925-939. [PUBMED](#) | [CROSSREF](#)
89. Correia DV, Lopes A, Silva-Santos B. Tumor cell recognition by $\gamma\delta$ T lymphocytes: T-cell receptor vs. NK-cell receptors. *Oncol Immunology* 2013;2:e22892. [PUBMED](#) | [CROSSREF](#)
90. Oppenheim DE, Roberts SJ, Clarke SL, Filler R, Lewis JM, Tigelaar RE, Girardi M, Hayday AC. Sustained localized expression of ligand for the activating NKG2D receptor impairs natural cytotoxicity in vivo and reduces tumor immunosurveillance. *Nat Immunol* 2005;6:928-937. [PUBMED](#) | [CROSSREF](#)
91. Hudspeth K, Silva-Santos B, Mavilio D. Natural cytotoxicity receptors: broader expression patterns and functions in innate and adaptive immune cells. *Front Immunol* 2013;4:69. [PUBMED](#) | [CROSSREF](#)
92. Siegers GM, Lamb LS Jr. Cytotoxic and regulatory properties of circulating V δ 1+ $\gamma\delta$ T cells: a new player on the cell therapy field? *Mol Ther* 2014;22:1416-1422. [PUBMED](#) | [CROSSREF](#)
93. Sanz M, Mann BT, Chitrakar A, Soriano-Sarabia N. Human V δ 2 T cells and their versatility for immunotherapeutic approaches. *Cells* 2022;11:3572. [PUBMED](#) | [CROSSREF](#)
94. Beckers D, Jainarayanan AK, Dustin ML, Capera J. T cell resistance: on the mechanisms of T cell non-activation. *Immune Netw* 2024;24:e42. [PUBMED](#) | [CROSSREF](#)
95. Dhatchinamoorthy K, Colbert JD, Rock KL. Cancer immune evasion through loss of MHC class I antigen presentation. *Front Immunol* 2021;12:636568. [PUBMED](#) | [CROSSREF](#)
96. Sari G, Rock KL. Tumor immune evasion through loss of MHC class-I antigen presentation. *Curr Opin Immunol* 2023;83:102329. [PUBMED](#) | [CROSSREF](#)
97. Revesz IA, Joyce P, Ebert LM, Prestidge CA. Effective $\gamma\delta$ T-cell clinical therapies: current limitations and future perspectives for cancer immunotherapy. *Clin Transl Immunology* 2024;13:e1492. [PUBMED](#) | [CROSSREF](#)
98. Costa GP, Mensurado S, Silva-Santos B. Therapeutic avenues for $\gamma\delta$ T cells in cancer. *J Immunother Cancer* 2023;11:e007955. [PUBMED](#) | [CROSSREF](#)
99. Hu Y, Hu Q, Li Y, Lu L, Xiang Z, Yin Z, Kabelitz D, Wu Y. $\gamma\delta$ T cells: origin and fate, subsets, diseases and immunotherapy. *Signal Transduct Target Ther* 2023;8:434. [PUBMED](#) | [CROSSREF](#)